MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON

TYPES 2N3742, 2N4926, 2N4927 AND

TX2N3742, TX2N4926, TX2N4927

1. SCOPE

1.1 Scope. - This specification covers the detail requirements for silicon, NPN, transistors in high-voltage, amplifier application in electronic-equipment circuits. (See 3.4 and 6.2 herein.) The prefix "TX" identifies devices meeting the special process-conditioning, testing, and screening requirements in 4.4 herein, and the stringent quality-control (LTPD) requirements in Tables I, II, and III herein.

1.2 Outline and dimensions. - See Figure 1 herein. (TO-39)

1.3 Maximum ratings. - (At T_A = + 25°C, unless otherwise specified):

	P_T 1/ at: T_A = +25°C	P_T 2/ at: T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_J	T_{stg}	Baro. Pressure (Alt.)
	W	W	Vdc	Vdc	Vdc	mAdc	°C	°C	ft
2N3742	1	5	300	300	7	50	+200	-65to+200	100,000
2N4926	1	5	200	200	7	50	+200	-65to+200	100,000
2N4927	1	5	250	250	7	50	+200	-65to+200	100,000

1/ For T_A > +25°C, derate linearly at 5.71 mW/°C.
2/ For T_C > +25°C, derate linearly at 28.6 mW/°C.

1.4 Particular electrical characteristics. - (At T_A = +25°C, unless otherwise specified):

ALL TRANSISTOR TYPES HEREIN													
h_{FE} 1/				h_{fe}		$\lvert h_{fe} \rvert$		V_{BE} (sat) 1/		V_{CE}(sat) 1/		C_{obo}	
at: I_C = 3mAdc V_{CE}=10Vdc		at: I_C = 30mAdc V_{CE} =10Vdc		at: f=1 kHz I_C = 10mA V_{CE} = 10V		at: f=20MHz I_C = 10mA V_{CE} = 20V		at: I_C=30mAdc I_B =3mAdc		at: I_C=30mAdc I_B=3mAdc		at: f≥0.1MHz ≤1.0MHz V_{CB}=20Vdc	
Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min	Max.	Min.	Max.
---	---	---	---	---	---	---	---	Vdc	Vdc	Vdc	Vdc	pf	pf
10	---	30	120	20	200	2.0	8.0	---	1.2	---	1.2	---	6

1/ Pulsed (see 4.3.1 herein).

FSC-5961

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-19500	Semiconductor Devices, General Specification For.

STANDARDS

MILITARY

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-750	Test Methods for Semiconductor Devices
MIL-STD-1276	Leads, Weldable, for Electronic Component Parts

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Requirements. - Requirements for the transistor shall be in accordance with Specification MIL-S-19500 and as otherwise specified herein.

3.2 Abbreviations and symbols. - The abbreviations and symbols used herein are defined in Specification MIL-S-19500.

3.3 Design and construction. - The transistor shall be of the design, construction, and physical dimensions specified on Figure 1.

3.3.1 Terminal arrangement. - The terminal arrangement on the transistor shall be as indicated in Figure 1.

3.3.2 Operating position. - The transistor shall be capable of proper operation in any position.

3.3.3 Terminal-lead material and finish. - The terminal-lead material shall be Type K (Kovar) or Type F (Alloy 52) conforming to requirements in MIL-STD-1276, except that a nickel under-plating of thickness: 20 microinches minimum, 100 microinches maximum, may be used. The terminal-lead finish shall be gold (per pertinent requirements in MIL-STD-1276); however, if so specified in the contract or order, the lead-material finish may be tin, or tin-coating over gold. This tin-finish requirement shall not be construed as affecting adversely the qualified-product status of the device, or applicable JAN marking. (See 6.3b herein.)

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3.3.3.1 Selectivity of terminal-lead material. - Where choice of lead material (see 3.3.3 above) is desired, stipulation for the particular material shall be made in contract or order. *(See 6.3a herein.)*

3.4 Performance characteristics. - The transistor performance characteristics shall be as specified in Tables I, II, and III herein. Except where specifically differentiated for respective transistor types (see 1.3, 1.4, and Tables I, II, and III herein), the performance requirements, including operating characteristics, ratings, test conditions, and test limits, apply equally to all transistor types covered herein.

3.4.1 "TX" special requirements. - Specified "TX" device-types shall, in addition to all other performance requirements herein, be process-conditioned, tested, and screened in accordance with 4.4 herein. (At option of the manufacturer, the non-TX device types may be subjected to the same process-conditioning and testing. (See 3.5.1 herein.)

3.5 Marking. - Except as otherwise specified herein, marking shall be in accordance with Specification MIL-S-19500. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the "classification" type designation only. The "manufacturer's identification" and "country of origin" may, at option of the manufacturer, be omitted from being marked directly on the semiconductor device covered herein.

3.5.1 "TX" identification marking. - Marking of devices conforming to "TX"and all other applicable requirements herein (see 3.4) shall include the prefix "TX" immediately preceding the type designation.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

4.2 Qualification and acceptance inspection. - Qualification and Quality Conformance inspection shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein. (See 4.2.2 herein.) Groups A, B, and C inspection shall consist of the examinations and tests specified in Tables I, II and III, respectively, herein. Quality Conformance Inspection shall include inspection of Preparation for Delivery (see 5.1 herein.)

4.2.1 Specified LTPD for subgroups. - The LTPD specified for a subgroup in Tables I, II, and III herein shall apply for all of the tests, combined, in the subgroup.

4.2.2 Qualification testing. - The non-TX device types shall be considered subject to Qualification tests. It shall be understood that procurement of the TX device types will be made from sources for the approved non-TX devices as listed in Qualified Products List (QPL)-19500. Upon request and validation by the qualifying activity, qualification may be extended (and appropriate QPL-listing effected) to TX devices manufactured by approved sources for the non-TX devices.

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4.2.2.1 Qualification relative to lead-material-finish. - Devices constructed with either of the lead-material finishes established herein (see 3.3.3) may be submitted for Qualification approval. The type of finish used by the manufacturer shall be stipulated in the submittal data for the product to be tested.

4.2.3 Special Group A criteria for Qualification inspection. - For Qualification inspection, only 10 failures will be permitted for all Group A tests combined. Hereto, where the manufacturer deems that the data in paragraph 4.4.2 of Specification MIL-S-19500 are invokable, notification shall be made to the Chief, Components and Materials Standardization Branch, Procurement and Production Directorate, Fort Monmouth, New Jersey 07703, Attention: AMSEL-PP-EM-2.

4.2.4 Group B-Group C life test samples. - Samples of non-TX transistor types, only, that have been subjected to Group B, 340-hour life test may be continued on test for 1000 hours in order to satisfy Group C life test requirements. These samples shall be predesignated, and shall remain subjected to the Group C 1000-hour evaluation after they have passed the Group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1000 hours shall be computed for 1000-hour acceptance criteria. (See 4.2.5 below). "TX" transistor types shall be subjected to 1000-hour life tests under Group B inspection requirements, and the lambda (λ) established thereto in Table II shall apply.

4.2.5 Group C testing. - Group C tests shall be performed on a lot every 6 months. (See Table III herein). The contractor shall, throughout the course of a contract or order, permit the Government representative to scrutinize all test data and findings covering manufacturer's test program on Group C characteristics and parameters for the product concerned. Upon determination by the Government inspector (in advance of Group C, 6-month, test results) that Group C parameters are not being adequately met, the Government inspector may require lot-by-lot inspection, normally for a minimum of 3 consecutive lots, to be performed for required Group C tests.

4.2.6 Disposition of sample units. - Sample units that have been subjected to Group B, Subgroup 2, 4, and 5 tests shall not be delivered on the contract or order. Sample units that have been subjected to and have passed Group B, Subgroups 1, 3, 6, 7, 8, and 9 tests and all Group C tests (these tests to be considered non-destructive), may be delivered on the contract or order provided that, after Group B and C inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

4.2.7 Copies of lot-inspection data. - When specified in the contract or order, one copy of the Quality Conformance inspection data, including manufacturer's test data and findings, covering the device inspection lot being shipped shall accompany the device(s) concerned.

4.3 Particular examination and test requirements. -

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4.3.1 "Pulsed"-conditions measurements. - Measurements required herein to be effected under pulsed conditions, shall be made in accordance with "Pulse Measurements" requirements in Section 4 of Standard MIL-STD-750.

4.3.2 Interval for End-Point Test Measurements. - All applicable End-Point Test measurements shall be performed within 96 hours after sample units have been subjected to required physical-mechanical or environmental test(s).

4.3.3 Mechanical damage resulting from tests. - Except for intentionally deforming, mutilating, or dismembering mechanical-stress tests to which samples are subjected, there shall be no evidence of mechanical damage to any sample unit as a result of any of the Groups A, B, or C tests.

4.3.4 Safe Operating Area (DC Operation) test. - Each transistor under test shall sustain the applicable test conditions and the following acceptance criteria shall apply:

a. I_C (for each transistor) shall not vary more than 10% during the dc operation; and

b. All other specified End-Point Test(s) limits shall not be exceeded, after the dc-operation test.

4.4 Process-conditioning, testing and screening for "TX" devices. - All "TX" devices (100% of each lot) shall be subjected to the process-conditioning, testing, and screening specified in 4.4.2 through 4.4.6.4 herein, in accordance with the procedural sequence established in Figure 2 herein, prior to submission of the lot to Group A, B, and C tests. The manufacturer shall permit the authorized Government representative to witness, concurrent with time of manufacturer's performance of these tests, the process-conditioning, testing, and screening on the devices. Those conditioning and screening tests normally performed by a manufacturer as standard-production tests, need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than those specified herein.

4.4.1 Quality Assurance records and determinants. - The manufacturer shall maintain and make available for perusal and review at any time, during a period of 3 years minimum, to the authorized Government representative or other contracting customer concerned, complete lot-by-lot records for these devices. All devices that fail to meet the process-conditioning, testing, and screening requirements specified herein shall be removed from the respective lot, and the test-failure identity and quantity removed shall be recorded in the permanent lot history. (See 6.3, 6.4, and 6.5 herein.)

4.4.2 High-temperature storage. - The transistors shall be stored for at least 24 hours at a minimum temperature (T_A) of +200°C.

4.4.3 Thermal shock (temperature cycling). - The transistors shall be subjected to temperature cycling (thermal shock) in accordance with MIL-STD-750, Method 1051, Test Condition C, except that cycling duration shall be 10 continuous cycles and exposure time at temperature extremes shall be 15 minutes minimum.

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4.4.4 Acceleration. - The transistors shall be subjected to the constant acceleration (centrifugal) test per Method 2006 in Standard MIL-STD-750, except that the following shall apply:

a. Centrifugal force - 20,000G
b. Orientation = Y1, only.
c. Hold time of 1 minute shall not apply hereto.

4.4.5 Seal (fine-leak) test. - The transistors shall be subjected to the Seal test per Standard MIL-STD-202, Method 112, Test Condition C, procedure IIIa or IIIb (see 4.4.5.1 or 4.4.5.2 below), except that testing for gross leaks shall be in accordance with 4.4.5.3 below.

4.4.5.1 Conditions for procedure IIIa. - The sealed chamber in which the transistors are placed shall be pressurized with helium gas to 50 psig, minimum, for a minimum of 4 hours. At end of that period, the transistors shall be removed from the chamber and, within 30 minutes, be subjected to a helium leak-detection test. Those transistors shwing a leakage rate of greater than 1 x 10^{-7} cc/sec of helium, when measured at a differential pressure of 1 atmosphere, shall be rejected and removed from the lot.

4.4.5.2 Conditions for procedure IIIb. - The devices shall be placed in an activation tank, pressurized with Krypton 85 tracer gas in a nitrogen solution, for sufficient time to detect a leak rate of 1 x 10^{-8} atmospheric cubic centimeters per second (atm cc/sec). Within four hours after subjection to this pressurization, the leak rate of the devices shall be determined on an attribute basis using the general equation shown below. Any device exhibiting a leak rate equal to or greater than 1 x 10^{-8} atm cc/sec shall be removed from the lot. The general equation for use with radioactive-gas leak test equipment is:

$$Q = \frac{R}{SKT(P_e 2 - P_i 2)}$$

where:

Q = leak rate in atm cc/sec.
R = net counting rate of tested part above background in cts/min.
S = specific activity of the test gas mixture in uCi/atm cc.
K = counting efficiency of the system for the given part in cts/min uCi.
P_e = pressure of test gas in activation tank during pressurization in atm abs.
P_i = pressure inside part under test in atm abs.
T = duration of pressurization in test gas mixture in seconds.

4.4.5.3 Gross leak test. - The transistors shall be immersed for a minimum period of 15 seconds in a suitable, non-corrosive liquid maintained at approximately +100°C, and shall be observed for emanating bubbles. Those transistors from which a bubble trace is observed shall be rejected and removed from the lot.

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4.4.6 Burn-in conditioning, tests, and evaluation (screening). - To enable proper evaluation of the transistors conditioned and tested in accordance with the following requirements, each transistor and lot shall be handled or identified in such manner that results for each individual transistor and lot can be ascertained and recorded.

4.4.6.1 Pre burn-in measurements. - The transistors shall be tested for the characteristics listed in Table IV below, and the values found shall be recorded. Those transistors that fail to meet the specified limits shall be rejected and removed from the lot.

TABLE IV. PRE BURN-IN MEASUREMENTS.

TEST METHOD PER MIL-STD-750	EXAMINATION OR TEST 1/	TEST CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN.	MAX.	
3036	Collector-base cutoff current:	Bias Cond. D				
	2N3742	V_{CB} = 250Vdc	I_{CBO}	---	500	nAdc
	2N4926	V_{CB} = 150Vdc	I_{CBO}	---	500	nAdc
	2N4927	V_{CB} = 200Vdc	I_{CBO}	---	500	nAdc
3076	Forward-current transfer ratio:	V_{CE} = 10 Vdc I_C = 30mAdc Pulsed; 2/	h_{FE}	30	120	---

1/ See 3.4 herein.
2/ See 4.3.1 herein.

4.4.6.2 Burn-in conditioning. - The transistors shall be energized and operated for 168 hours minimum, under the following conditions:

T_A = +25°C (without heat sink)

P_T = 1 W

2N3742: V_{CB} = 200 Vdc
2N4926: V_{CB} = 100 Vdc
2N4927: V_{CB} = 100 Vdc

4.4.6.3 Post burn-in tests and measurements. - After burn-in conditioning, the transistors shall be permitted to return to room-ambient temperature thermal-equilibrium and retested, within a maximum total elapsed time of 96 hours, for the characteristics listed in Table IV above. The values now found for each individual transistor shall not exceed the following (respectively applicable to the pre burn-in measurements per Table IV above):

ΔI_{CBO} = 100% or 50 nanoamperes, whichever is greater.

Δh_{FE} = ± 20%.

4.4.6.4 Burn-in conditioning evaluation (screening). - Those transistors whose measured characteristics exceed the delta (Δ) limits specified in 4.4.6.3, above, or the limits of Table IV, shall be rejected and removed from the lot. Where 10% or more of the transistors subjected to the Burn-in test are found to exceed those limits, the entire lot shall be considered non-acceptable as "TX" devices.

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TABLE I. GROUP A INSPECTION.

TEST METHOD PER MIL-STD-750	EXAMINATION OR TEST 1/	CONDITIONS	LTPD NON-TX	LTPD TX	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Subgroup 1		10	5				
2071	Visual and mechanical examination	---			---	---	---	---
	Subgroup 2		5	3				
3001	Collector-base breakdown voltage:	Bias Cond.D I_C=100 uAdc						
	2N3742				BV_{CBO}	300	---	Vdc
	2N4926				BV_{CBO}	200	---	Vdc
	2N4927				BV_{CBO}	250	---	Vdc
3011	Collector-emitter breakdown voltage	Bias Cond.D I_C = 10mAdc Pulsed; 2/						
	2N3742				BV_{CEO}	300	---	Vdc
	2N4926				BV_{CEO}	200	---	Vdc
	2N4927				BV_{CEO}	250	---	Vdc
3026	Emitter-base breakdown voltage	Bias Cond. D I_E = 100uAdc			BV_{EBO}	7	---	Vdc
3036	Collector-base cutoff current:	Bias Cond. D I_E = 0						
	2N3742	V_{CB} = 250Vdc			I_{CBO}	---	500	nAdc
	2N4926	V_{CB} = 150Vdc			I_{CBO}	---	500	nAdc
	2N4927	V_{CB} = 200Vdc			I_{CBO}	---	500	nAdc
3061	Emitter-base cutoff current	Bias Cond. D V_{EB} = 5 Vdc			I_{EBO}	---	200	nAdc
	Subgroup 3		7	5				
3066	Base-emitter saturation voltage	Test Cond. A Pulsed; 2/ I_C = 10mAdc I_B = 1 mAdc			V_{BE} (sat)	---	1.0	Vdc
3066	Base-emitter saturation voltage	Test Cond. A Pulsed; 2/ I_C = 30mAdc I_B = 3 mAdc			V_{BE} (sat)	---	1.2	Vdc

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TABLE I. GROUP A INSPECTION (Continued)

TEST METHOD PER MIL-STD-750	EXAMINATION OR TEST 1/	CONDITIONS	LTPD NON-TX	LTPD TX	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Subgroup 3 - (Continued)							
3071	Collector-emitter saturation voltage	Pulsed; 2/ I_C = 10mAdc I_B = 1 mAdc			V_{CE} (sat)	---	1.0	Vdc
3071	Collector-emitter saturation voltage	Pulsed; 2/ I_C = 30mAdc I_B = 3 mAdc			V_{CE} (sat)	---	1.2	Vdc
3076	Forward-current transfer ratio	Pulsed; 2/ I_C = 3.0mAdc V_{CE} = 10Vdc			h_{FE}	10	---	---
3076	Forward-current transfer ratio	Pulsed; 2/ I_C = 10mAdc V_{CE} = 10Vdc			h_{FE}	15	---	---
3076	Forward-current transfer ratio	Pulsed; 2/ I_C = 30mAdc V_{CE} = 10Vdc			h_{FE}	30	120	---
3076	Forward-current transfer ratio	Pulsed; 2/ I_C = 50mAdc V_{CE} = 20Vdc			h_{FE}	20	---	---
	Subgroup 4		10	7				
3306	Magnitude of common-emitter small-signal short-circuit transfer ratio	I_C = 10mAdc V_{CE} = 20Vdc f = 20 MHz			$\lvert h_{fe} \rvert$	2	8	---
3206	Small-signal short-circuit forward-current transfer ratio	I_C = 10mAdc V_{CE} = 10Vdc			h_{fe}	20	200	---
3236	Output capacitance (open circuit)	V_{CB} = 20Vdc I_E = 0 f $\geq$ 0.1 $\leq$ 1.0 MHz			C_{obo}	---	6	pf

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TABLE I. GROUP A INSPECTION. - (Continued)

TEST METHOD PER MIL-STD-750	EXAMINATION OR TEST 1/	CONDITIONS	LTPD		SYMBOL	LIMITS		UNIT
			NON-TX	TX		MIN.	MAX.	
	Subgroup 4 -(Cont'd)							
3240	Input capacitance (Output open-circuited)	V_{EB} = 1Vdc I_C = 0 $f \geq 0.1 \leq 1.0$ MHz			C_{ibo}	8	80	pf

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TABLE II. GROUP B INSPECTION.

TEST METHOD PER MIL-STD-750	EXAMINATION OR TEST 1/	CONDITIONS	LTPD NON-TX	LTPD TX	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Subgroup 1		20	20				
2066	Physical dimensions	---			---	---	---	---
	Subgroup 2		15	15				
2026	Solderability	---			---	---	---	---
1051	Temperature cycling	Test Cond. C			---	---	---	---
1056	Thermal shock (glass strain)	Test Cond. A			---	---	---	---
2/	Seal (leak rate)	Test Cond. C, Procedure III; Test Cond. A for gross leaks			---	---	1×10^{-7}	atm cc/sec
1021	Moisture resistance	---						
	End Point Tests:							
3036	Collector-base cutoff current: 2N3742 2N4926 2N4927	Bias Cond. D $I_E = 0$ $V_{CB} = 250$Vdc $V_{CB} = 150$Vdc $V_{CB} = 200$Vdc			 I_{CBO} I_{CBO} I_{CBO}	 --- --- ---	 500 500 500	 nAdc nAdc nAdc
3076	Forward-current transfer ratio	Pulsed; 3/ $I_C = 30$mAdc $V_{CE} = 10$Vdc			h_{FE}	30	120	---
	Subgroup 3		15	15				
2016	Shock	Non-operating 1500G, 5 blows of 0.5 msec ea. in orientations X1, Y1, Y2, Z1 (total = 20 blows)			---	---	---	---

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TABLE II. GROUP B INSPECTION - (Cont'd).

TEST METHOD PER MIL-STD-750	EXAMINATION OR TEST 1/	CONDITIONS	LTPD		SYMBOL	LIMITS		UNIT
			NON TX	TX		MIN.	MAX.	
	Subgroup 3 (Cont'd)							
2046	Vibration fatigue	Non-operating			---	---	---	---
2056	Vibration, variable frequency	---			---	---	---	---
2006	Constant acceleration	20,000G Orientations X1, Y1, Y2, Z1			---	---	---	---
	End Point Tests:							
	Same as for Subgroup 2 above							
	Subgroup 4		20	20				
2036	Terminal strength (lead fatigue)	Test Cond. E			---	---	---	---
	Subgroup 5		20	20				
1041	Salt atmosphere (corrosion)	---			---	---	---	---
	Subgroup 6		10	10				
4/	High -temperature operation:	T_A = +150°C 5/						
3036	Collector-base cutoff current:	Bias Cond. D I_E = 0						
	2N3742	V_{CB}=250Vdc			I_{CBO}	---	100	uAdc
	2N4926	V_{CB}=150Vdc			I_{CBO}	---	100	uAdc
	2N4927	V_{CB}=200Vdc			I_{CBO}	---	100	uAdc
---	Low -Temperature Operation:	T_A =-55°C 5/						
3076	Forward-current transfer ratio	Pulsed; 3/ I_C=30mAdc V_{CE}=10Vdc			h_{FE}	15	---	---

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TABLE II. GROUP B INSPECTION -(Cont'd).

TEST METHOD PER MIL-STD-750	EXAMINATION OR TEST 1/	CONDITIONS	LTPD NON-TX	LTPD TX	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Subgroup 7		10	10				
---	Safe operating area (dc operation):	Test Circuit and procedure per Fig. 3 herein; 6/ T_A = +25°C t = 1 sec. 1 cycle			---	---	---	---
	Test #1	I_C = 50mAdc V_{CE} = 20Vdc						
	Test #2	I_C = 10mAdc V_{CE} = 100Vdc						
	Test #3 2N3742	I_C = 3.3mAdc V_{CE} = 300Vdc						
	2N4926	I_C = 5 mAdc V_{CE} = 200 Vdc						
	2N4927	I_C = 4 mAdc V_{CE} = 250Vdc						
	End Point Tests:							
	Same as for Subgroup 2 above							
	Subgroup 8		7	λ=7				
1031	High-temperature life (non-operating)	T_{stg} = +200°C t = 340 hrs 7/			---	---	---	---
	End Point Tests:							
3036	Collector-base cutoff current:	Bias Cond. D I_E = 0						
	2N3742	V_{CB} = 250Vdc			ΔI_{CBO}	---	8/	9/
	2N4926	V_{CB} = 150Vdc			ΔI_{CBO}	---	8/	9/
	2N4927	V_{CB} = 200Vdc			ΔI_{CBO}	---	8/	9/

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TABLE II. GROUP B INSPECTION -(Cont'd)

TEST METHOD PER MIL-STD-750	EXAMINATION OR TEST 1/	CONDITIONS	LTPD NON-TX	LTPD TX	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Subgroup 8 (Continued)							
	End Point Tests: (Continued)							
3076	Forward-current transfer ratio	Pulsed; 3/ I_C = 30mAdc V_{CE} = 10Vdc			Δh_{FE}	---	±25%	9/
	Subgroup 9		7	λ=7				
1026	Steady state operation life:	T_A = +25°C P_T = 1 W 7/			---	---	---	---
	2N3742	V_{CB} = 200Vdc						
	2N4926	V_{CB} = 100Vdc						
	2N4927	V_{CB} = 100Vdc						
	End Point Tests:							
	Same as for Subgroup 8 above							

1/ See 3.4 and 4.3.2 herein.

2/ Per Method 112 in Standard MIL-STD-202.

3/ See 4.3.1 herein.

4/ In this Subgroup, the sample units subjected to the High-Temp. Oper. Test should be permitted to return to and be stabilized at room ambient temperature prior to their being subjected to the Low Temp. Oper. test.

5/ Test measurement(s) shall be made after thermal equilibrium has been reached at the temperature specified.

6/ See 4.3.4 herein.

7/ See 4.2.4 herein.

8/ ± 100% or 100 nA, whichever is greater.

9/ Change from Initial Group A reading.

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TABLE III. GROUP C INSPECTION. 1/

TEST METHOD PER MIL-STD-750	EXAMINATION OR TEST 2/	CONDITIONS	LTPD NON-TX	LTPD TX	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Subgroup 1		15	15				
1001	Barometric pressure, reduced (altitude operation)	Pressure=8mmHg Normal mounting t = 60 sec., minimum			---	---	---	---
	Measurement during test:							
3036	Collector-base cutoff current:	Bias Cond. D I_E = 0						
	2N3742	V_{CB} = 300Vdc			I_{CBO}	---	100	uAdc
	2N4926	V_{CB} = 200Vdc			I_{CBO}	---	100	uAdc
	2N4927	V_{CB} = 250Vdc			I_{CBO}	---	100	uAdc
3201	Small-signal short-circuit input impedance	V_{CE} = 10Vdc I_E = 10mAdc f = 1 kHz			h_{ie}	75	750	ohms
3211	Small-signal open-circuit reverse-voltage transfer ratio	V_{CE} = 10Vdc I_C = 10mAdc f = 1 kHz			h_{re}	---	$1x10^{-4}$	--
3216	Small-signal open-circuit output admittance	V_{CE} = 10Vdc I_C = 10mAdc f = 1 kHz			h_{oe}	5	50	umhos
	Subgroup 2		λ=10	---				
1031	High-temperature life (non-operating)	T_{stg}=+200°C t = 1000 hrs. 3/						
	End Point Tests:							
3036	Collector-base cutoff current	Bias Cond. D I_E = 0						
	2N3742	V_{CB} = 250Vdc			I_{CBO}	---	4/	5/
	2N4926	V_{CB} = 150Vdc			I_{CBO}	---	4/	5/
	2N4927	V_{CB} = 200Vdc			I_{CBO}	---	4/	5/

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TABLE III. GROUP C INSPECTION. 1/-(Cont'd).

TEST METHOD PER MIL-STD-750	EXAMINATION OR TEST 2/	CONDITIONS	LTPD NON-TX	LTPD TX	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Subgroup 2 End Point Tests (Continued)							
3076	Forward-current transfer	Pulsed; 6/ I_C = 30mAdc V_{CE} = 10Vdc			Δh_{FE}	---	±25%	5/
	Subgroup 3		λ=10	---				
1026	Steady state operation life:	T_A = +25°C P_T = 1 W 3/						
	2N3742	V_{CB} = 200Vdc						
	2N4926	V_{CB} = 100Vdc						
	2N4927	V_{CB} = 100Vdc						
	End Point Tests:							
	Same as for Subgroup 2 above							

1/ See 4.2.5 herein.

2/ See 3.4 and 4.3.2 herein.

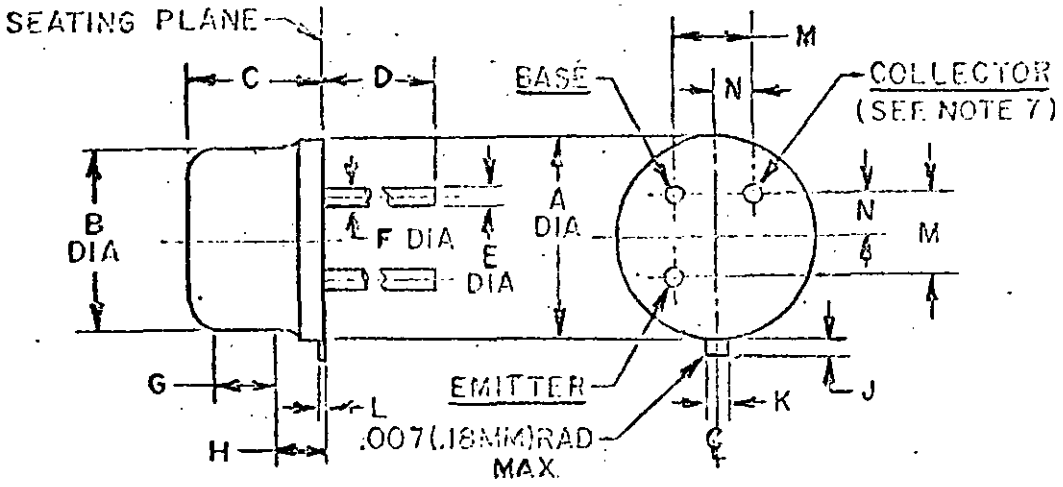
3/ See 4.2.4 herein.

4/ ± 100% or 100 nA, whichever is greater.

5/ Change from Initial Group A reading.

6/ See 4.3.1 herein.

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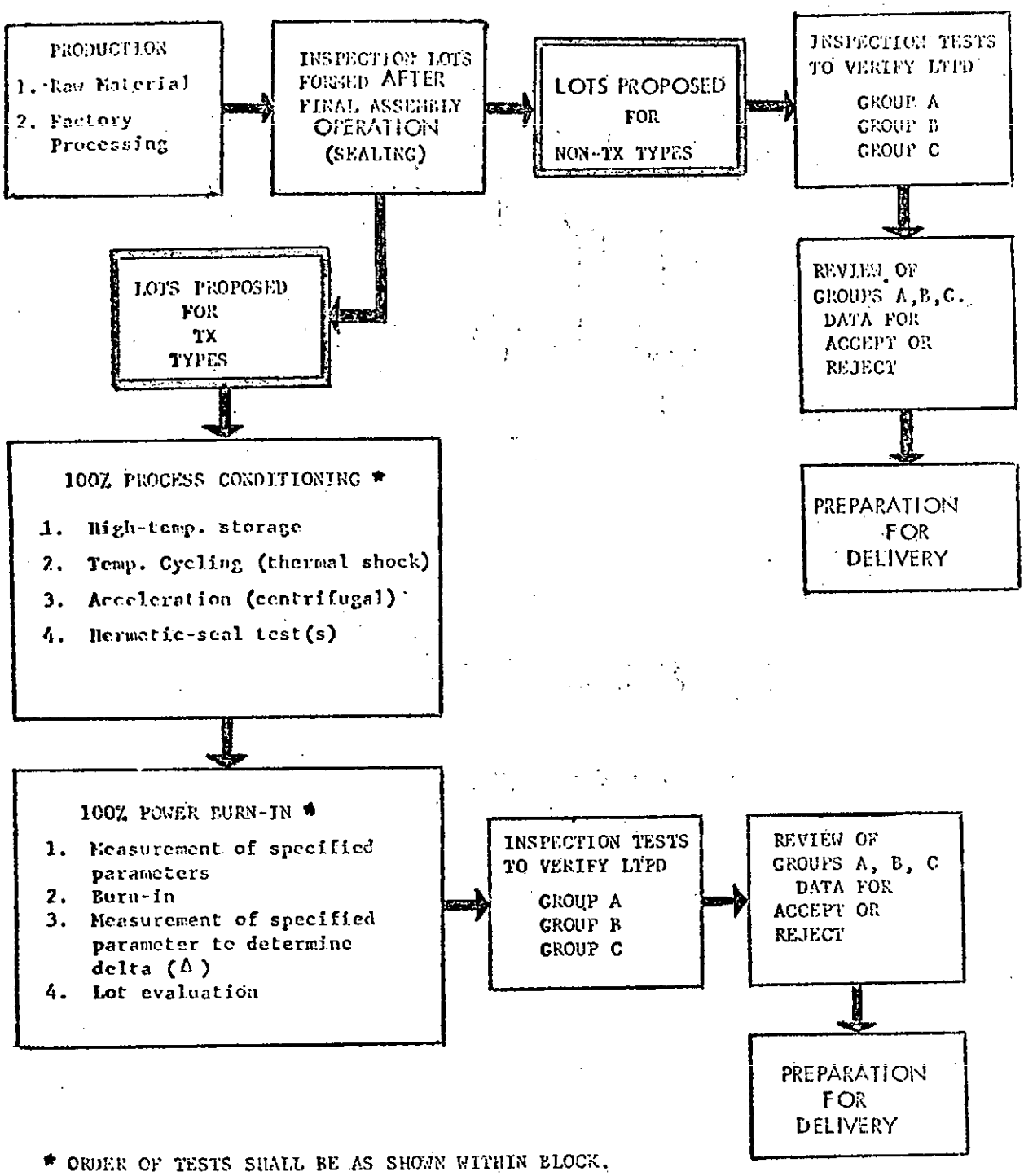
NOTES:

1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plans.
3. Measured in the zone .050 (1.27mm) and .250 (6.35 mm) from the seating plane.
4. Variations on Dim B in this zone shall not exceed .010 (.25 mm).
5. Outline in this zone is not controlled.
6. When measured in a gaging plane .054 + .001, -.000(1.37+.03,-.00 mm) below the seating plane of the transistor max dia leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller dia leads shall fall within the outline of the max dia lead tolerance. Figure 2 preferred measured method.
7. The collector shall be internally connected to the case.
8. Measured from the maximum diameter of the actual device.
9. All 3 leads.

LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.335	.370	8.51	9.40	
B	.305	.335	7.75	8.51	
C	.240	.260	6.10	6.60	
D	.500	.750	12.70	19.05	9
E	.016	.021	.41	.53	2,9
F	.016	.019	.41	.48	3,9
G	.100	---	2.54	---	4
H	---	---	---	---	5
J	.029	.045	.74	1.14	8
K	.028	.034	.71	.86	
L	.009	.125	.23	3.18	
M	.1414 Nom		3.59 Nom		6
N	.0707 Nom		1.80 Nom		6

FIGURE 1. OUTLINE AND DIMENSIONS.

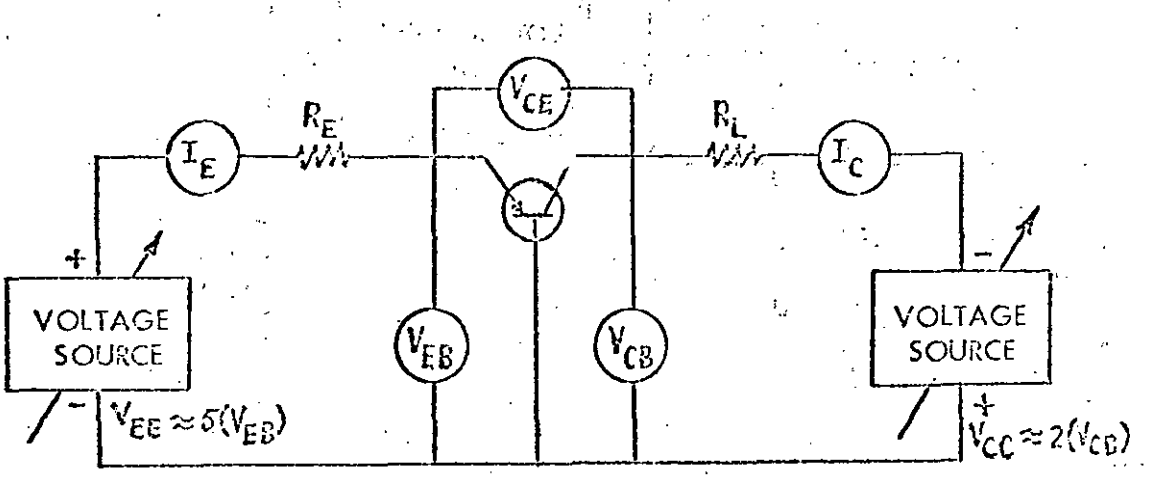
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* ORDER OF TESTS SHALL BE AS SHOWN WITHIN BLOCK.

FIGURE 2. FLOW CHART OF NON-TX AND TX TYPES

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TEST PROCEDURE:

a. Starting with V_{CC} and V_{EE} at a low value, increase V_{CC} to obtain approximately the specified V_{CE}; increase V_{EE} to obtain approximately the specified I_C. Increase V_{CC} and subsequently adjust V_{EE} to obtain the specified V_{CE} and I_C. Operate the transistor at specified case temp; for the specified time duration.

b. At end of operation time: decrease V_{CC} to obtain V_{CE} near zero; turn off V_{EE}; turn off V_{CC}; remove transistor-under-test.

Fig. 3. Safe Operating Area (DC Operation) test circuit.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. - Preparation for delivery, and the inspection of preparation for delivery, shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes. - The notes included in Specification MIL-S-19500, with the following additions or exceptions, are applicable to this document.

6.2 Application guidance. - To insure proper circuit application, particular attention should be given to the differential voltage and gain characteristics, and ratings, pertinent to the individual transistor types covered herein.

6.3 Ordering data. -

a. Terminal-lead material: see 3.3.3.1 herein.
b. Terminal-lead finish: see 3.3.3 herein.
c. Lot-inspection data: see 4.2.7 herein.

6.4 Procurement of "TX" types. - See 4.2.2 herein.

6.5 Quality assurance for "TX" types.- It is expected that where the permanent lot record, maintained as required in 4.4.1 herein, begins to show a condition of failures in excess of the LTPD rate specified herein, or a condition of entire-lot rejection (see 4.4.6.4 herein) is recorded, the Government procuring activity will determine any additional monitoring controls and evaluation criteria needed to attain restoration of the prescribed (herein) quality assurance for acceptance of product.

6.6 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Commanding General, U. S. Army Electronics Command, Fort Monmouth, New Jersey, Attention: AMSEL-PP-EM-2.

CUSTODIAN:

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Preparing Activity:

ARMY-EL

Project No. 5961-A174

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-19500/405(EL) SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON TYPES 2N3742, 2N4926, 2N4927

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

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